

House Price Prediction Project Summary

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In this project, I worked on a house price prediction dataset to understand how different property features affect the selling price of a house. I started by loading and exploring the dataset, checking for missing values, duplicate records, and understanding the available features. After cleaning the data, I converted categorical columns into numerical format so that machine learning models could process them effectively. The dataset was then divided into training and testing sets using an 80:20 ratio. To predict house prices, I trained two machine learning models: Linear Regression and Random Forest Regressor. The performance of both models was evaluated using MAE, RMSE, and R^2 Score. Based on the results, the Random Forest model performed better and produced more accurate predictions. From the analysis, I observed that factors such as area, number of bedrooms, number of bathrooms, and preferred location had a strong influence on house prices. One interesting finding was that houses with similar sizes sometimes had very different prices because of location and additional amenities. Overall, this project helped me understand the complete machine learning workflow, including data preprocessing, model training, evaluation, and visualization. For real estate businesses, focusing on location, property size, and important facilities can help make better pricing decisions and improve market competitiveness.